

FIRST FIT COMMENTS

```
Activities Terminal Mar 17 21:11 manishaa@localhost:~$ # FIRST FIT (First Fit) Page Replacement Algorithm
# First Fit looks for the first frame (from the beginning)
# that is free. If no free frame is available, it replaces
# the page in the first frame.
# Simple but not always optimal.

# Steps:
1. Start from the first frame.
2. If a free frame is found, load the page into it.
3. If no free frame is available, replace the page in
   the first frame (index 0).
4. Continue for all pages in the reference string.

# Example:
# Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3
manishaa@localhost:~$
```

```
Activities Terminal Mar 17 21:11 manishaa@localhost:~$ # FIRST FIT (First Fit) - Example

# Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3

# Step-by-step (First Fit):
# Ref: 1 2 3 4 1 2 1 2 3 4 5
# Frames:
# [ 1 1 1 4 4 4 5 5 5 3 3 ]
# [ - 2 2 2 2 2 2 1 1 4 4 ]
# [ - - 3 3 3 3 2 2 2 2 5 ]

# Page Faults = 9

manishaa@localhost:~$
```

```
Activities Terminal Mar 17 21:11 manishaa@localhost:~$ # FIRST FIT - How it works (Detailed)

# 1. 1 -> Load into first frame (F1)
# 2. 2 -> Load into first free frame (F2)
# 3. 3 -> Load into first free frame (F3)
# 4. 4 -> No free frame, replace first frame (F1)
# 5. 1 -> No free frame, replace first frame (F1)
# 6. 2 -> No free frame, replace first frame (F1)
# 7. 5 -> No free frame, replace first frame (F1)
# 8. 1 -> No free frame, replace first frame (F1)
# 9. 2 -> No free frame, replace first frame (F1)
# 10. 3 -> No free frame, replace first frame (F1)
# 11. 4 -> No free frame, replace first frame (F1)
# 12. 5 -> No free frame, replace first frame (F1)

manishaa@localhost:~$
```

```
Activities Terminal Mar 17 21:11 manishaa@localhost:~$ # FIRST FIT - Summary

# Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3
# Page Faults: 9

# Final frame content after all references:
# F1: 5
# F2: 4
# F3: 2

# Note:
# First Fit is simple but may not give the least number
# of page faults compared to other algorithms like LRU
# or Optimal.
manishaa@localhost:~$
```